



**FINAL EXAMINATION
SEMESTER II SESSION 2022/2023**

FACULTY OF INFORMATICS AND COMPUTING

DIPLOMA PROGRAMME

COURSE : DATABASE
COURSE CODE : ITD10503
DURATION : 2 HOURS AND 30 MINUTES

INSTRUCTIONS TO CANDIDATES

1. Answer ALL questions.
2. Answer Section A and B in the Answer Booklet provided. Begin each answer on a new page.

This Question Booklet consists of 4 printed pages.

Section A

Question 1 [25 Marks]

a) Discuss the **THREE (3)** limitations involve in Figure 1.

[6 Marks]

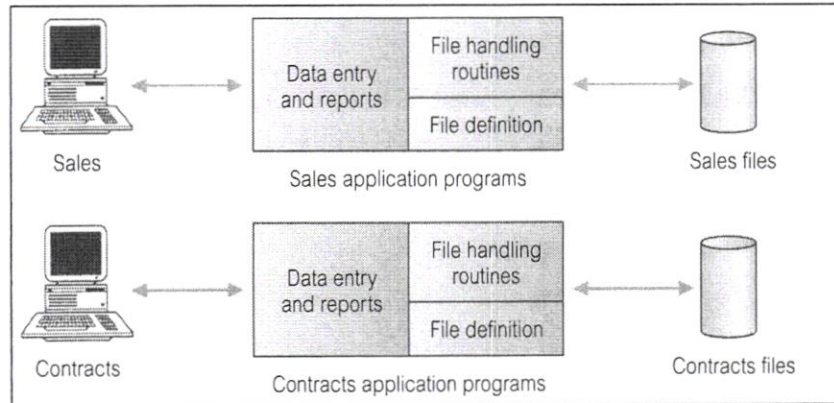


Figure 1. File based processing.

b) Suggest **ONE (1)** approach to eliminates limitations in (a).

[1 Mark]

c) State **THREE (3)** advantages of implementing suggestion approach in (b).

[6 Marks]

d) Discuss **THREE (3)** purposes of ANSI-SPARC Three-level architecture as shown in Figure 2.

[6 Marks]

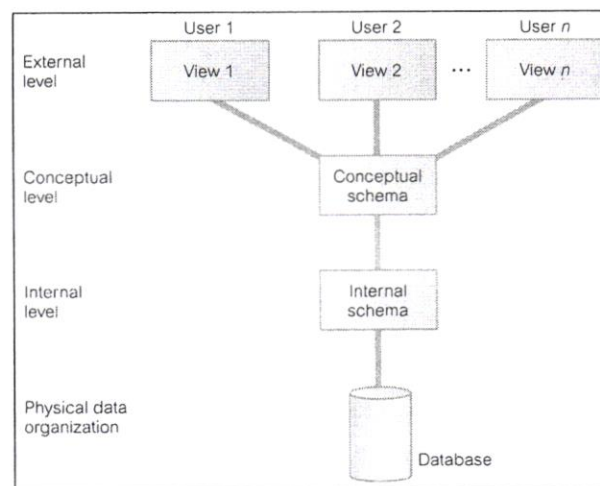


Figure 2. ANSI-SPARC Three-level architecture

e) Briefly discuss **THREE (3)** benefits of implementing views to an organization.

[6 Marks]

Question 2 [25 Marks]

a) Given the following business rules, create the appropriate Chen Model for each of the specified relationship. **[8 Marks]**

- A company operates four departments.
- Each department employs employees.
- Each of the employees may or may not have one or more dependents.
- Each employee may or may not have an employment history.

b) Define Candidate key and Referential integrity. **[4 Marks]**

c) The relationship structure is portrayed in two simple and common formats. Briefly describe these two formats. **[4 Marks]**

d) Convert the above ER diagram in Figure 3 into a relational database schema. Indicate primary keys and referential integrity constraints in the schema. **[9 Marks]**

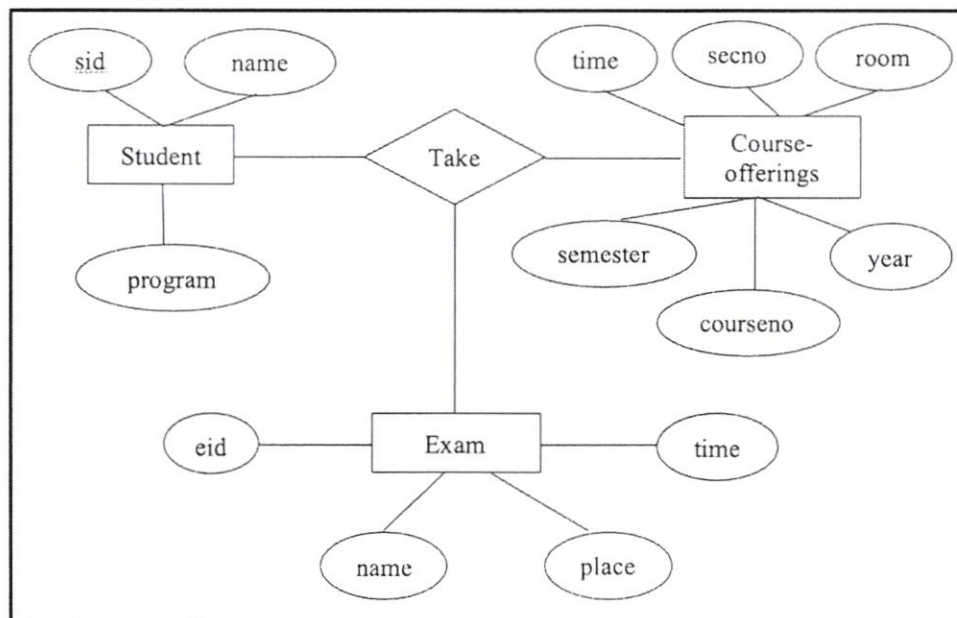


Figure 3. Entity Relationship Diagram

Section B [50 Marks]

- a) Iwhizz is a wholesale company that purchases computers from **suppliers** and sells them to **customers**. Previously, all sales records, computer stocks, and **supplier** lists were recorded manually using **file-based system**. Therefore, Iwhizz wishes to upgrade the file system to a more systematic system. Delima Solutions, a company that offers system **development services**, has been tasked with upgrading the **file-based system**. To ensure that the **development** of this new system runs smoothly, several phases of **database life cycle** need to be carried out. As a **database designer**, **discuss those phases**.

[20 Marks]

- b) Analyze the following description of Iwhizz Computer's business rules. Draw the Entity Relationship Diagram (ERD). Show the primary keys and completeness constraint in your ERD.

Iwhizz has a number of **suppliers** and each supplier supply many products. A **product** can be supply by more than one supplier. Iwhizz also keeps record of suppliers who are not currently supplying any products. Supplier details are **supplier code, name and address**. When suppliers deliver the products, Iwhizz's staff will record the **product details** which are **code, description, quantity and price per unit**. These products have been grouped according to several product groups. **Product groups** also have codes and descriptions.

Iwhizz has huge number of customers. The customer sends an order for the product. **Orders** have a **code, order date and quantity**. Customers can place as **many orders** as possible or not place orders at all. **Each order** is made by only **one customer**. Customer code, name, address is recorded for each customer. **Each customer** order must contain at least **one product**.

Iwhizz have several staff. Recorded staff data such as staff code, name, IC no, address and telephone no. The staff manage customer orders.

[30 Marks]**END OF QUESTION PAPER**